INTERNAL HOM OF BOUNDED-TYPE COHERENT COMPLEXES ON A LOCALLY NOETHERIAN SCHEME IS COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Inputs lemma-identify-pseudo-coherent-noetherian and lemma-quasi-coherence-internal-hom are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Allowed standard prerequisites (editorial).

- Affine reduction and pseudo-coherence identification. lemma-identify-pseudo-coherent-noetherian (same chapter, NOT carded, declared external). On affines, D^-_Coh objects are pseudo-coherent; reduces to the algebra problem.
- Quasi-coherence of internal Hom. lemma-quasi-coherence-internal-hom (same chapter, NOT carded, declared external). Translates R SheafHom(K, L) into R Hom_A(K, L) over affines.
- Spectral sequences for Ext. Injectives, remarks spectral-sequences-ext and variant (cross-chapter, named). Two convergent spectral sequences decompose Ext^p(K, H^q(L)) => Ext^{p+q}(K, L).
- Finiteness of Ext between finite modules. Algebra, lemma ext-noetherian (cross-chapter, named). Ext^p_A(M, N) finite for finite A-modules over Noetherian A.

Source reference. Original reference 08YR: focused-seeds/geometry/S9-perf2-coherent-internal-hom/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G1Y: focused-seeds/geometry/S9-perf2-coherent-internal-hom/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G20: focused-seeds/geometry/S9-perf2-coherent-internal-hom/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E8: sources/stacks-perfect/pinned-original-source.tex, source lines 2480–2480. Navigation pointer only: the counted proof

Date: stacks-project:perfect.tex:local-snapshot-20261012; GU FDL 1.2 or later.

cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A6H: sources/stacks-perfect/pinned-original-source.tex, source lines 2657–2657. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a locally Noetherian scheme. If L is in $D^+_{Coh}(\mathcal{O}_X)$ and K in $D^-_{Coh}(\mathcal{O}_X)$, then $R\mathcal{H}om(K, L)$ is in $D^+_{Coh}(\mathcal{O}_X)$.*

Proof. It suffices to prove this when X is the spectrum of a Noetherian ring A. By Lemma 08E8 we see that K is pseudo-coherent. Then we can use Lemma 0A6H to translate the problem into the following algebra problem: for $L \in D^+_{Coh}(A)$ and K in $D^-_{Coh}(A)$, then $R\operatorname{Hom}_A(K, L)$ is in $D^+_{Coh}(A)$. Since L is bounded below and K is bounded above there is a convergent spectral sequence

$$\operatorname{Ext}^p_A(K, H^q(L)) \Rightarrow \operatorname{Ext}^{p+q}_A(K, L)$$

and there are convergent spectral sequences

$$\operatorname{Ext}^i_A(H^{-j}(K), H^q(L)) \Rightarrow \operatorname{Ext}^{i+j}_A(K, H^q(L))$$

See Injectives, Remarks 0G1Y and 0G20. This finishes the proof as the modules $\operatorname{Ext}^p_A(M, N)$ are finite for finite A-modules M, N by Algebra, Lemma 08YR. □